

Green Compliance vs. Productivity Premium: Evaluating the Environmental-TFP Link in Indonesia

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ABSTRACT

This study examines how adopting green practices relates to the productivity of manufacturing firms in Indonesia. Using the 2021 Indonesian Annual Manufacturing Survey (IBS), from which we analyse a final sample of 10,573 firms, we apply a two-stage approach. First, we estimate each firm's Total Factor Productivity (TFP). Second, we test how this productivity relates to a "Green Index", which measures whether a firm uses eco-friendly raw materials, produces green goods, and holds environmental certifications like ISO 14001. To ensure the reliability of our estimates, we employ alternative productivity measures and cluster standard errors at the industry level. The results show that adopting green practices does not significantly increase a firm's productivity, either as a combined index or as individual actions. Instead, we find that productivity is strongly driven by foreign ownership and the use of technologies like automation and artificial intelligence. This pattern is consistent with a literature suggesting that green adoption in developing-country manufacturing is shaped more by regulatory compliance than by efficiency-seeking, though our data cannot establish the underlying motive. We therefore interpret our results as showing no detectable short-term productivity premium from green practices, rather than as evidence of any particular motive.

1 INTRODUCTION

As environmental concerns grow worldwide, manufacturing firms face increasing pressure to adopt eco-friendly practices. While this transition is essential, business leaders and policymakers question whether it makes good business sense. Some argue that environmental standards can spark innovation and improve efficiency. However, adopting green raw materials or obtaining environmental certifications also requires significant upfront costs that may erode short-term profits. In a developing economy like Indonesia, where manufacturing is a major engine of growth, whether going green boosts firm productivity remains an open question.

To address this, we use the 2021 Indone-

sian Annual Manufacturing Survey, which covers more than 30,000 establishments, and analyse a final sample of 10,573 firms to examine the link between eco-friendly practices and firm productivity. We investigate whether using green inputs, producing green goods, or holding environmental certifications is associated with higher efficiency. The data reveal no detectable productivity premium from these green practices. Instead, productivity is most strongly associated with foreign ownership, innovation, and modern technologies such as automation and AI. We are careful to frame these as conditional associations rather than causal effects, given the cross-sectional design.

2 LITERATURE REVIEW

2.1 Green Practices and Firm Performance

Studies on how green initiatives affect performance often show mixed results because of the trade-off between higher revenues and compliance costs. He et al. (2015) found that adopting ISO 14001 certification among Chinese firms did not significantly improve financial performance, as the boost in exports and sales was offset by an almost equal increase in operating costs. This suggests that in the early stages of going green, transition costs tend to cancel out potential efficiency gains.

2.2 Environmental Shocks and Technology Adoption

The adoption of new technology is often triggered by external pressures such as energy price shocks. Calì et al. (2022) found that higher fuel prices in developing countries led to increased productivity for factories in Indonesia, as firms replaced inefficient fuel-powered machines with electricity-based technology. This highlights that environmental pressures can catalyze modernization, a strong driver of plant productivity.

2.3 Foreign Exposure and Input Learning

Firm productivity in developing countries is shaped by integration with international networks. Amiti and Konings (2007), using Indonesian census data, showed that reducing tariffs on imported inputs delivered a much larger productivity boost than reducing tariffs on final goods, through learning, variety, and quality effects. This supports the role of foreign affiliation and global supply chains in overcoming the inefficiencies of purely domestic firms.

2.4 Regulatory Pressure as a Motivation for Green Practices

In developing countries, eco-friendly adoption is often driven more by legal compliance

than by market-efficiency incentives. A recent study of the Indonesian automotive industry (Setiawan et al., 2026) finds the green transition is heavily shaped by institutional pressure and government regulation. This perspective suggests that holding an environmental certification may often reflect compliance obligations rather than a deliberate efficiency strategy.

2.5 Empirical Evidence of Green Performance in Domestic Manufacturing

Overall, green initiatives in Indonesian manufacturing have not yet shown a positive correlation with performance. Hendrawan and Suhartini (2025) find that green process and product innovations in basic-industry and chemical firms can have a negative and significant impact on performance, as large initial investments are not yet matched by market willingness to pay for greener products. This provides structural context for why transition costs may currently neutralize efficiency gains.

3 RESEARCH QUESTIONS

- Is using eco-friendly raw materials, producing green goods, or holding environmental certifications associated with higher firm efficiency?
- Are green initiatives associated with productivity, or are other factors such as foreign ownership and automation more strongly related?

4 DATA AND METHODS

4.1 Data Source

This study uses firm-level data from the 2021 Annual Manufacturing Survey (IBS) collected by Statistics Indonesia (BPS), covering medium and large manufacturing establishments. Because the data are from a single year, we conduct a cross-sectional analysis.

Table 1: Variable Descriptions and Data Types

Variable	IBS Code	Description	Type
Stage 1: Productivity Estimation			
Gross output (Y)	YPRVCU	Total production value	Continuous
Value added (VA)	VTLCU	Output value minus cost of intermediate inputs (used for robustness check)	Continuous
Labour (L)	LTLNOU	Total number of workers	Continuous
Capital (K)	V1115	Book value of fixed assets at the end of 2021	Continuous
Materials (M)	(RDNVCU + RIMVCU)	Domestic and imported raw materials	Continuous
Stage 2: Green Index Components			
Green input	GREEN (R219)	Uses eco-friendly raw materials	Dummy (1/0)
Green output	YGREEN (R221)	Produces eco-friendly goods	Dummy (1/0)
Green certification	SNIISO / ECOLAB	Holds ISO 14001 or carries an eco-label	Dummy (1/0)
Firm Controls & Fixed Effects			
Foreign ownership	DASING (R210b)	Foreign ownership share greater than 0%	Dummy (1/0)
R&D	RND (R216)	Firm conducts in-house R&D	Dummy (1/0)
Innovation	INOV (R218)	Introduced a new product or process	Dummy (1/0)
Automation / AI	AUTOAI (R227)	Uses robotics, AI, or automation	Dummy (1/0)
Industry / Province	DISIC5_3 / DPROVI	2-digit industry and province identifiers	Fixed effects

Source: IBS 2021, BPS.

4.2 Sample Composition and Selectivity

Figure 1 shows the distribution of firms across two-digit industries in the analysis sample. The food industry dominates with 3,332

firms, followed by non-metallic minerals, rubber and plastics, and textiles. The remaining 14 industries are grouped as “Other.” The sector is diverse but heavily concentrated in food production.

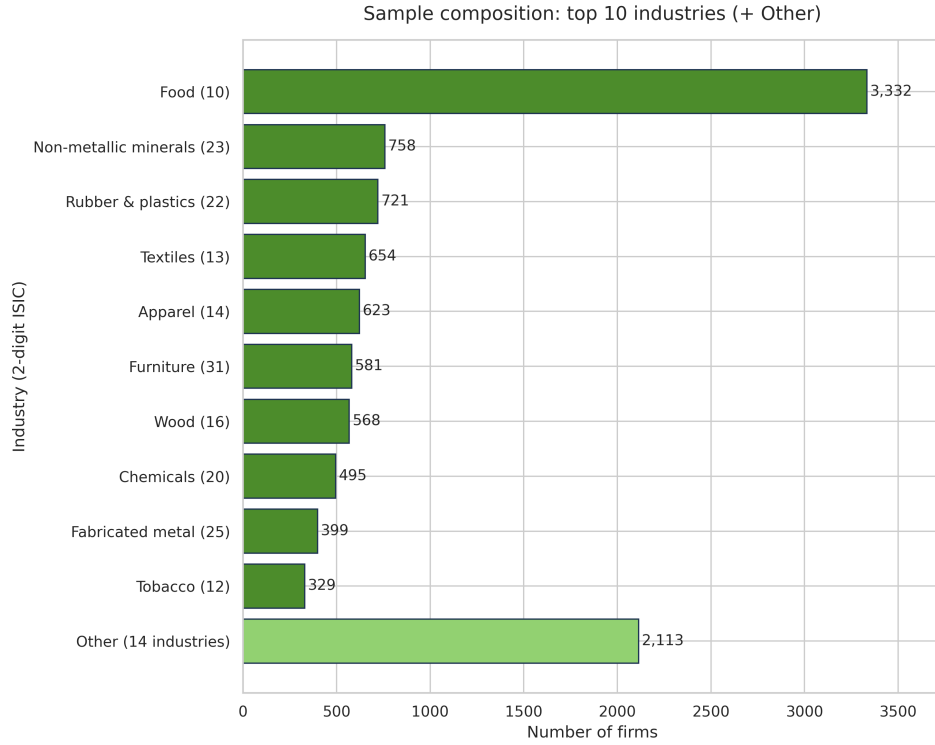


Figure 1: Sample Composition: Top 10 Industries and Others

Our final sample is 10,573 firms, obtained after cleaning the 30,788 surveyed establishments. We retain only firms that responded to the green module (16,619 firms, since we do not impute non-response), then drop firms with non-positive values for output, value

added, labour, capital, or materials because the production function is estimated in logs (leaving 14,623 firms), and finally drop observations with missing values on any Stage 2 regressor. The final sample spans 24 two-digit industries and 34 provinces.

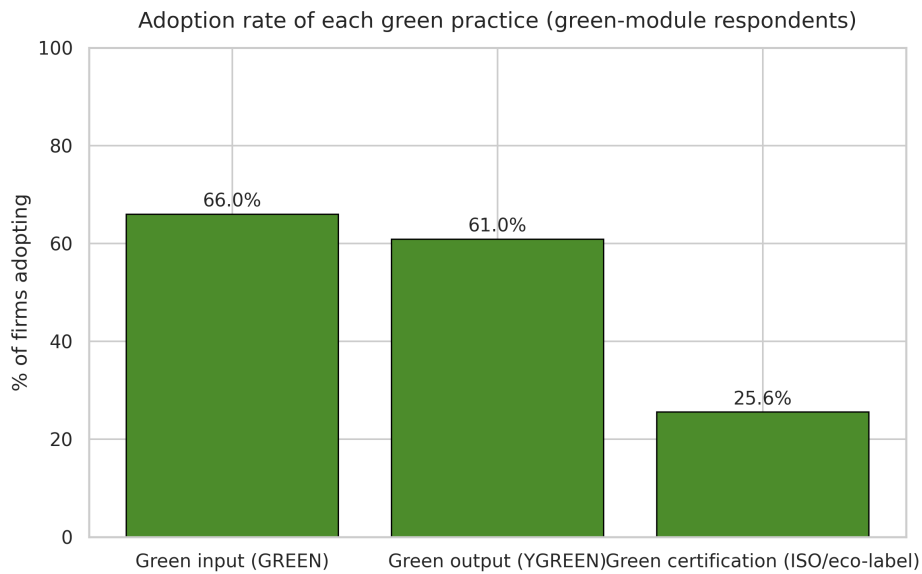


Figure 2: Adoption Rate of Each Green Practice (green-module respondents)

Figure 2 reports adoption rates within this respondent sample. Among respondents, 66.0% use eco-friendly inputs and 61.0% produce green goods, but only 25.6% hold a formal certification such as ISO 14001 or an eco-label. The gap suggests that while many firms adopt greener materials or products, securing internationally recognized certification remains a hurdle, likely due to administrative and financial costs. These rates apply to the respondent sample only.

4.3 Empirical Strategy

We use a two-stage approach. Because TFP is unobserved, we estimate it (Stage 1) before relating it to the Green Index (Stage 2). All monetary values are log-transformed and extreme values are winsorized at the 1st and 99th percentiles on the log scale.

We estimate the production function by OLS. We acknowledge that OLS input elasticities are subject to transmission bias. This bias occurs because firms choose inputs partly in response to productivity shocks they observe but we do not, ultimately biasing the estimated coefficients. Control-function corrections (Olley-Pakes, 1996; Levinsohn-Petrin, 2003; Akerberg, Caves and Frazer, 2015) are not feasible here because they require a panel dimension the IBS cross-section lacks. We therefore treat our TFP estimates as a transparent but imperfect benchmark, and emphasize that the Stage 2 conclusions concern the relative association between green practices and productivity, not the absolute level of TFP.

4.4 Stage 1: Estimating Productivity

Our main specification uses a gross-output Cobb–Douglas function with labour, capital, and materials:

$$\ln Y_i = \beta_0 + \beta_1 \ln L_i + \beta_2 \ln K_i + \beta_3 \ln M_i + \varepsilon_i \quad (1)$$

Firm-level TFP is the estimated residual (the Solow residual). The estimated elasticities are $L = 0.37$, $K = 0.13$, $M = 0.61$, implying returns to scale of about 1.11 (close to constant returns and economically plausible). As a robustness check we also estimate a value-added specification with labour and capital only.

4.5 The Green Index

We measure environmental effort with an additive Green Index from 0 to 3: one point each for using green inputs, producing green goods, and holding a certification. We exclude a general “green technology” item whose survey responses were too vague to code reliably.

4.6 Stage 2: Testing the Green-TFP Link

We relate estimated TFP to the Green Index:

$$\ln TFP_i = \alpha_0 + \alpha_1 \text{GreenIndex}_i + X_i' \eta + \gamma_s + \delta_p + u_i \quad (2)$$

Here α_1 is the coefficient of interest. X_i' includes firm controls (foreign ownership, R&D, innovation, automation); γ_s and δ_p are industry and province fixed effects. Standard errors are clustered by 2-digit industry; with only 24 clusters they may be modestly understated, which bears on the significant control variables but not on the insignificant result for green practices. We also test the components separately. Because the data are a single cross-section, results are conditional associations, not causal effects.

5 RESULTS

Table 2: OLS Regression Result: Association Between Green Practices and TFP

	Gross Output TFP (Index)	VA TFP (Index)	Gross-Output TFP (Components)
	(1)	(2)	(3)
GreenIndex	0.001 (0.007)	−0.012 (0.017)	
GREEN			−0.003 (0.025)
YGREEN			−0.010 (0.026)
Green Cert			0.030 (0.021)
DASING	0.104** (0.048)	0.341*** (0.084)	0.101** (0.048)
RND	0.043 (0.041)	0.125** (0.058)	0.041 (0.040)
INOV	0.070** (0.028)	0.108*** (0.036)	0.069** (0.028)
AUTOAI	0.090** (0.040)	0.119** (0.050)	0.088** (0.040)

Note: Standard errors clustered by 2-digit industry in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: IBS 2021, processed by the authors.

Table 2 reports the Stage 2 estimates. The first model, our main gross-output specification, shows that the Green Index has a coefficient of 0.001 that is statistically insignificant. This indicates that adopting green practices has no detectable association with a firm’s productivity. By contrast, factors such as foreign ownership, general innovation, and the use of automation or AI show positive and highly significant effect on efficiency. The strong positive effect of foreign ownership (DASING) particularly highlights the importance of global exposure. As noted in our literature review, foreign-affiliated firms tend to benefit from faster technology transfer and better access to high-quality international supply chains, which help them overcome domestic market inefficiencies.

To ensure these findings are robust, the second model re-estimates the relationship using value-added TFP. Here, the Green Index remains insignificant with a coefficient of −0.012. Although the value-added specifica-

tion is statistically less reliable due to an inflated returns to scale, it confirms that the lack of a productivity premium from green practices is consistent across different estimations. Furthermore, the control variables in the second model retain their positive signs but are larger in magnitude. This is expected because the gross-output specification already accounts for material inputs, which is the primary channel where foreign ownership and innovation advantages operate. Finally, since R&D is only significant in the value-added model and not in our main specification, we do not emphasize it as a robust driver of productivity.

6 MODEL LIMITATIONS AND ROBUSTNESS

Several limitations apply. Because the data are a single cross-section, our estimates are associations, not causal effects. This concern extends to the controls, since more pro-

ductive firms may attract foreign investment or adopt automation rather than the reverse. Transmission bias is also visible in our own estimates, as the value-added labour elasticity (≈ 0.97) and returns to scale (≈ 1.27) exceed plausible priors. This is why we adopt the better-behaved gross-output specification (returns to scale ≈ 1.11) as our main measure and read TFP as a relative ranking rather than a structural parameter. Finally, the green module is observed only for a non-random subset skewed toward smaller firms, so the null result may not generalize to the larger establishments absent from the sample. Clarifying why the module covered only part of the sample requires BPS documentation and is left to future work. Panel data and control-function methods would address both the bias and the coverage issues.

Within these bounds, the findings are robust. The null green result holds across gross-output and value-added TFP, across the combined index and its individual components, and under standard errors clustered by industry to allow for shared shocks among similar firms.

7 CONCLUSION

This study finds no evidence that adopting green practices is associated with a short-term productivity premium for Indonesian manufacturing firms. Neither the combined Green Index nor its individual components show a significant relationship with TFP, whereas foreign ownership, innovation, and automation are positively associated with it. Our findings are narrower than a claim about motives; conditional on industry, province, and firm controls, green practices simply show no detectable link to productivity. One plausible explanation, supported by prior work on Indonesian manufacturing (Setiawan et al., 2026; Hendrawan and Suhartini, 2025), is that green adoption at this stage is driven largely by regulatory compliance, with high transition costs neutralizing short-term gains.

The implications are tentative. If transition costs currently outweigh immediate ben-

efits, financial incentives, tax relief, or technical assistance could help firms offset the burden of going green. For managers, treating certification as a compliance checklist is unlikely on its own to improve competitiveness. Instead, pairing green investment with broader technological upgrades, such as automation (which our estimates associate positively with productivity), may be more promising.

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